SmartAlarm — Physical device compatibility catalog



Concept illustration — proposed integration, not an installed system.

Research date: **2026-10-23**. Status: **document-verified candidates; no hardware integration, procurement, or field acceptance completed**. This catalog supplements WF-03, WF-14, WF-15, and WF-17. The Vietnamese and English editions contain the same scope and conclusions.

Current design baseline: 3.0 · BMS integration: 24 October 2026. Original research/requirement dates above are preserved.

1. Deployment decision

Use manufacturer-supported interfaces to observe existing fire panels, pump controllers, and releasing panels. Keep their local automatic protection functions operational independently of SmartAlarm. A protocol connection, a writable BACnet property, a pump-running signal, and confirmed water discharge are four different capabilities. None alone demonstrates that SmartAlarm may safely initiate suppression or that water reaches the protected hazard.

The six candidates below are alternatives or complementary components, not a single certified assembly. Availability, exact firmware, regional approval, optional parts, and permitted actions require project-specific vendor confirmation. A manufacturer's listing of its own component does not certify the complete SmartAlarm installation.

2. Six concrete candidates

ID / product binding	Documented interface and feedback	Command boundary and unresolved items	Proposed SmartAlarm use
PHY-01 — Siemens Cerberus PRO FC922 / FC924; FCA2034-A1 S2 or FCA2035-A1 S3	FC922/FC924 are identified in the panel datasheet A6V10349710. The US accessory datasheet A6V10349679 describes S2 BACnet as monitoring-only; S3 retains monitoring-only BACnet while adding separate remote-viewer control.	Do not equate remote-viewer control with BACnet actuation. Obtain regional SKU, firmware, license scope, PICS/object export, and supported event states. No suppression-release capability is established here.	Read event/state data through a site edge adapter; preserve panel/point identifiers and communication quality.
PHY-02 — Honeywell NOTIFIER BACNET-GW-3 with NFS-320 / NFS2-640 / NFS2-3030 and appropriate NFN interface	DN-6877:C, 2014-08-04: BACnet/IP event objects; up to 14 NFN nodes and 15,000 objects; NCM/HS-NCM connection and browser configuration.	The listing C6, 2019-02-21 restricts use to supplementary functions and excludes triggering mass-notification announcements. Confirm current supply/support, firmware, chassis, power, and exact panel compatibility.	Supplementary monitoring; never treat this gateway as a replacement for the primary alarm or as a proven suppression command path.
PHY-03 — Simplex BACpac 4100-6111 with 4100ES / 4100U	S4100-0051 Rev.6, 10/2020, pp.1–4: 15,000-point variant; separate 4100-6038 RS-232 module; BACnet IP/MS/TP or Modbus TCP/RTU. Supplementary status interface.	Its PICS includes WriteProperty support; this does not establish an authorized panel-release command. Obtain the actual point mapping and permitted write semantics. Confirm software and separately ordered hardware.	Read-only status adapter for a specifically bound panel configuration.
PHY-04 — Tornatech GPX electric fire-pump controller, V2 communication profile	The GPx-V2 Modbus bulletin maps motor-run, voltage/current, pressure, and fault feedback. It also names remote/manual and automatic-start entries.	That short map does not fully specify write functions, enable conditions, persistence, or retry behavior. Do not automatically apply its V2 map to V2.1/current firmware. Exact controller suffix, options, and current register contract remain required.	Pump health and demand monitoring; command capability stays disabled until vendor-supported, project-approved validation.
PHY-05 — Tornatech GPD diesel fire-pump controller, V2 communication profile	The separate GPD-V2 Modbus bulletin maps engine-run, batteries, pressure, and diesel faults, plus named start entries.	GPX and GPD maps are not interchangeable. Confirm firmware, engine interface, installed sensors, writable registers, and automatic/manual mode behavior. Missing sensors must remain unknown.	Diesel readiness and pump-state monitoring with a dedicated adapter profile.
PHY-06 — Potter PFC-4410G3, stock 3006735	5401600 Rev.B, 4/23: pre-action/deluge releasing panel; alarm, trouble, supervisory, and waterflow/discharge-confirmation relays; Ethernet programming; proprietary RS-485 P-Link.	Ethernet/P-Link does not prove a public release API. Confirm output ratings, chosen program, compatible solenoid, feedback wiring, and optional relay modules. Do not assume releasing output equals waterflow confirmation.	Read approved relay indications through a supervised interface; local releasing panel owns suppression sequencing.

No additional interface license is established for PHY-02–06 by this review; this is not a claim that all configurations or engineering tools are free or included. Obtain a complete bill of materials and license statement.

3. Version and lifecycle findings that change implementation

Siemens regional distinction. The 2018 Siemens FC72x catalog describes FCA2034-A1 with basic and FCA2035-A1 with extended third-party BACnet commands. This differs from the US accessory wording above. Therefore the adapter key must include regional panel family and firmware, not just “Cerberus PRO” or the license label. The exact commands available at a particular site remain unverified.

NOTIFIER network constraints. Manual LS10014-000NF-E:D2, 2019-04-30, p.9 specifies a static IP and no NAT support. Proposed topology: keep the BACnet client in the permitted site network and export normalized events outward; do not assume a cloud VPN/NAT arrangement is supported.

Real pump start interfaces exist. Tornatech's current GPX manual V2026.09 r0, p.16 documents remote manual-start contact closure and remote automatic/deluge-start contact opening. These are physical controller inputs, not evidence of a cloud command API. Its GPX product page also identifies remote alarm contacts. Obtain the job-specific wiring diagram, contact ratings, supervision, and fail-state behavior before designing an I/O interface.

The GPD manual V2026.09 r0, pp.13–15 separately documents contact-based remote starts. Automatic stopping is not enabled by default and requires authority-having-jurisdiction authorization; commissioning requires an authorized provider. SmartAlarm must not introduce an unvalidated remote-stop function.

Select the current releasing panel deliberately. Potter's official releasing-panel catalog marks PFC-4410RC obsolete. The G3 is the candidate for new procurement, subject to vendor confirmation. Its product page explicitly identifies email monitoring as a non-UL-listed feature; it is not the safety event transport. Use the G3 manual 5403751 Rev.G, 11/25 and the manufacturer's compatible releasing-device list to request confirmation of the exact panel/solenoid/valve assembly. No valve pairing is approved by this catalog.

4. Proposed integration contract — SmartAlarm design, not manufacturer claims

For each installation, bind `manufacturer + regional_model + controller_serial + firmware + gateway_model + gateway_firmware + license + point_map_revision`. Store every observed point with source address, semantic type, engineering unit, source timestamp where available, receive timestamp, quality, and communication status. A point-map revision must be reviewable and rollback-capable.

Proposed semantic	Required evidence	Must not be inferred
`fire_alarm_active`	Valid panel point/state mapping	Camera confidence alone is not a panel alarm
`pump_start_demand`	Documented demand contact/register	Motor actually running
`pump_running`	Valid controller run feedback	Adequate pressure, flow, or extinguishment
`system_pressure`	Installed transducer, unit/scaling, calibration	Flow at a specific protected zone
`release_output_active`	Documented releasing circuit/output feedback	Valve movement or water discharge
`waterflow_confirmed`	Commissioned waterflow/pressure feedback and zone binding	Fire extinguished or every nozzle operating
`communication_lost`	Link supervision and timeout	Normal, safe, pump stopped, or failed actuation

The default adapter capability is `observe`. A potential `request_start` or other command requires a distinct signed capability record describing authority, exact operation, conditions, local interlocks, TTL, duplicate handling, feedback, and recovery. An uncertain response stays `unknown`; reconnection must not replay expired physical requests. Do not add reset/silence/disable/stop actions merely because a protocol supports writing.

5. Vendor RFI before procurement

RFI	Required written answer/evidence
RFI-01 — Identity and lifecycle	Regional SKU, orderable status, end-of-support date, firmware matrix, substitutions, and spare-part availability
RFI-02 — Complete package	Gateway/modules, licenses, engineering tools, power/backup, enclosure, isolation, protection, and authorized commissioning scope
RFI-03 — Read contract	Current PICS/register map, supported event/state types, polling/subscription limits, address conventions, scaling, sequence/time behavior, and communication-loss semantics
RFI-04 — Command contract	Explicit allowed/prohibited operations; write function/property; enabling configuration; physical input semantics; local priority; TTL/replay behavior; confirmation and rollback limits
RFI-05 — Physical assembly	Approved valve/solenoid/controller combinations, electrical loads, hydraulic design responsibility, pressure/flow sensor placement, cause-and-effect matrix, and supervision
RFI-06 — Regional acceptance	Applicable adopted codes and editions, component listings and limitations, system designer approval, AHJ/site acceptance requirements; no blanket international-compliance assertion
RFI-07 — Cybersecurity/support	Network segmentation, credentials, supported updates, vulnerability process, maintenance access, logging, outage behavior, and recovery procedures

6. Commissioning evidence and release gate

The following are proposed SmartAlarm acceptance requirements, not a replacement for manufacturer or local mandatory tests.

1. Record serial numbers, firmware, license proof, approved drawings, point map, and baseline controller configuration. Obtain approval of the complete cause-and-effect matrix.
2. In a controlled test arrangement, verify each mapped alarm, supervisory, fault, pump demand/run, pressure, releasing-output, and waterflow indication against the physical system. Record expected versus observed states and timestamps.
3. Test broken links, adapter restart, stale data, duplicate events, power/backup changes, and reconnect. Verify that local automatic protection continues and no stale event creates an unintended command.
4. If any command is authorized, test its approved preconditions, denial paths, local override, feedback loss, and recovery with the qualified site team. Do not perform live discharge merely to validate software connectivity.
5. Obtain signed factory/site acceptance records, calibration records, operator training, maintenance ownership, and an approved impairment/restoration procedure. Mark the adapter `field_accepted` only for that exact configuration.

Procurement recommendation: first bind to the installed panel/controller family, then buy the supported interface and required options. For a greenfield water-suppression installation, have the fire-protection designer specify the physical hydraulic and releasing system before selecting the SmartAlarm adapter. No purchase or vendor contact has been made.

7. Evidence register and access limits

All sources below were reviewed on **2026-10-23**. Document dates are publisher revision dates, not retrieval dates. Online presence does not establish current stock or support. Only manufacturer sources support product claims; no reseller claim is used.

Evidence	Publisher document / date	Access and scope
E01	Siemens A6V10349710; A6V10349679	Official indexed title/accessory passages; direct PDF retrieval was blocked. Revision must be obtained with the quotation.
E02	Siemens FC72x catalog, ©2018	Official indexed license passage; full download exceeded browser size limit. Regional distinction only; not a current command contract.
E03	NOTIFIER DN-6877:C, 2014-08-04	Official indexed datasheet passages.
E04	NOTIFIER listing C6, 2019-02-21; manual D2, 2019-04-30	Official indexed limitation/configuration passages; direct retrieval unavailable. Request controlled copies and current revisions.
E05	Simplex S4100-0051 Rev.6, 10/2020	Official six-page PDF read; selection and PICS tables examined.
E06	Tornatech GPx-V2; GPD-V2	Official two-page maps read; V2 identity verified, no printed revision date established. URL upload path is not a revision guarantee.
E07	Tornatech GPX; GPD, both V2026.09 r0	Official manuals read, particularly starting/stopping and commissioning; current manuals do not by themselves validate the older V2 register maps.
E08	Potter 5401600 Rev.B, 4/23; 5403751 Rev.G, 11/25	Official datasheet and manual retrieved; interfaces and revision verified.
E09	Potter G3 product page; releasing-panel catalog; device compatibility list	Live manufacturer pages reviewed; compatibility-list link retrieved but subsequent fetch timed out, so no specific valve/solenoid pairing is asserted.

Open evidence gaps remain procurement gates, not proof of incompatibility. This research demonstrates feasible integration paths and specific restrictions; it does not report completed interoperability or certify a deployment.

UX2-PHYSICAL · Visual supervision and physical feedback

The v2 console uses a pipe/valve/pump mimic, zone coverage and explicit device state. A requested command, gateway acknowledgement and observed physical effect are separate states. Unknown or stale feedback must use a labeled pattern and timestamp, never a green success state. Review-only actions in the prototype do not operate equipment. Local fire protection authority, interlocks and commissioning requirements in this document remain binding.

Verify F08–F20 with communication loss, conflicting local state and delayed feedback. Operator access to the schematic must not bypass command authorization. PA playback feedback is not proof of audibility to every occupant. Drone/robot task acceptance is not proof of arrival or rescue.

See UI/UX design, screen specifications and redesign audit.

8. Physical devices through BMS — implementation and qualification

A BMS is an observation/coordination path to physical systems, not a replacement for controller qualification. Register both native device identity and the BMS mirror path; a panel read directly and through BMS is still one physical source. F01/F07/F08 show source path, time, quality and mapping revision; BMS03 retains lineage across alarm paths.

Vendor profiles incorporated into the existing catalogue

Profile	Product and interface	Evidence and qualification
`bms.siemens.desigo-wsi`	Desigo CC REST / WSI subscriptions	V8 specification describes REST read/write, events and history. WSI authentication has version-specific session/heartbeat behavior. Verified northbound OPC is DA, not UA. Qualify project/extension, HTTPS and PICS for a BACnet binding. Siemens WSI
`bms.schneider.ebo-ews`	EcoStruxure Building Operation SOAP EWS / SmartConnector REST	The SmartConnector2.6/EWS1.2 guide was tested against EBO7.1.2. EBO3+ uses SoapEwsRestProvider. Do not assume root discovery or bulk support. Acknowledgement targets an alarm event, not every alarm-state transition. EBO EWS
`bms.jci.metasys-rest`	Metasys Version-pinned REST / SSE	API v6 documents objects, command schemas, alarms, histories and SSE. Inspect each batch result rather than HTTP200 alone. Monitoring and Commanding needs separate licensing; the Metasys name alone does not establish compatibility. Metasys API
`bms.honeywell.ebi-api`	Enterprise Buildings Integrator Vendor SDK / web services	EBI2025 advertises enterprise web services, but the public page does not provide an implementation contract. OIder R600 guide specifications discuss OPC DA/A&E/HDA and APIs; they do not establish equivalent REST or UA capability in EBI2025. EBI
`bms.honeywell.webs-niagara`	WEBs N4 Niagara OEM station interface	WEBs is Niagara-based and needs an OEM profile. Modules, station exports, permissions and licensing determine capabilities. Do not reuse EBI APIs. WEBs N4
`bms.tridium.niagara-local`	Niagara local station / Supervisor Qualified oBIX / licensed module	Supervisor4.6 documentation and the 2025 driver catalog identify oBIX. A marketplace REST module is a separate capability, not universal built-in REST. Alarm acknowledgement requires a qualified contract. Niagara local
`bms.tridium.nds-cloud`	Niagara Data Service Cloud API	NDS is a separate cloud service for histories/tags, live read/write and alarm queries. Prerequisites include SMA and a reseller. Alarm acknowledgement is unverified; WAN cloud is not a critical local-control fallback. Niagara Data Service
`bms.delta.enteliweb`	enteliWEB Vendor API / BACnet Web Services	The 4.23 catalog, updated November2022, lists an -API add-on. Confirm current entitlement; a brochure is insufficient for authentication, endpoints and lifecycle implementation. enteliWEB
`bms.alc.webctrl-events`	WebCTRL event/telemetry integration Outbound GET/POST / MQTT add-on	The 2025 sheet mentions alarm REST and outbound alarm-triggered GET/POST; MQTT is an add-on. Outbound events do not prove REST point commands or native acknowledgement. Initial scope is event/telemetry intake. WebCTRL

Profile	Product and interface	Evidence and qualification
`bms.alc.webctrl-soap-sdk`	WebCTRL SOAP / SDK SOAP / licensed Java SDK	SOAP needs a version-specific method contract; JavaSDK licensing is case-by-case. Qualify read, acknowledgement, commands and history by method and account scope. WebCTRL SDK
`bms.distech.ecnet-niagara`	EC-Net Facilities / EC-Net4 Niagara OEM station interface	EC-Net Facilities is the renamed Niagara-based EC-Net4. ECLYPSE REST belongs to the controller family, not automatically the EC-Net Supervisor. Qualify OEM modules and station access. EC-Net

Building-services boundaries

Subsystem	Required mapped data	Control and acceptance boundary
Fire panel / releasing	Alarm/trouble/supervisory, occurrence, zone, inhibit, local state	A BMS mirror does not grant reset/silence/release authority. Only qualified interfaces/models and approved cause-and-effect may be considered for commands.
Pumps/valves /tanks/flow	Run contact, pressure, valve limits, tank level, waterflow, quality/source time	Run/valve-open does not prove adequate flow. Use independent observations and expected effects; unknown remains unknown.
HVAC / smoke control	Fan command versus run, damper command versus limit, smoke zone, override	Comfort HVAC and smoke-control modes have different authority. Do not operate fans from an alarm rule without site-engineer approval.
Doors / access / lifts	Door contact, lock state, accessibility, lift availability/recall	Door-open does not prove a safe route; lift availability does not establish fire evacuation suitability. Never bypass controller interlocks.
PA / exit signage	Zone/group, emergency priority, device report, playback/fault	BMS signals are qualified rule triggers, not substitutes for PA supervision or intelligibility acceptance.

Required commissioning bundle for each BMS path

Record physical-controller manufacturer/model/firmware, BMS product/build, interface/provider/license, endpoint/principal scope, raw address and applicable byte order, units/enums/status mappings, topology, update semantics, latency/freshness budgets, read-only fixtures, any command allowlist, local priority/override, expected feedback and test permits. F17 stores bundle hashes/revisions and sign-off. Research JSON profiles are design-only; runtime admission requires evidence for this exact combination. Brochures or BTL listings do not establish per-method command permission.